

Sunlytics: Global Solar Energy Insights



1 Project Planning

1.1 Project Overview

This project analyzes the global relationship between **solar energy**, **economic development**, and **climate change** across 66 countries from 2000 to 2024. Data was collected from Our World in Data (OWID) and the World Bank and processed using Python, SQL, and data warehousing techniques. The objective is to understand solar energy trends and calculate the **Global Solar Trend Index (GSTI)**.

1.2 Project Objectives

- Collect and integrate global solar, economic, and environmental data.
- Clean, merge, and standardize datasets using Python.
- Build a **Data Warehouse (DW)** with fact and dimension tables.
- Perform analytical SQL queries on solar performance and CO₂ emissions.
- Forecast global solar electricity production up to 2035.
- Analyze correlations between solar energy, climate, and economy.
- Compute the **Global Solar Trend Index (GSTI)**.

1.3 Scope of Work

1. Solar energy generation and consumption.
2. CO₂ emissions and climate indicators.
3. Economic indicators including income and population.
4. Solar forecasting using ARIMA.
5. Comparative country analysis.

1.4 Project Methodology

1. **Data Collection:** Data gathered from OWID and World Bank for 66 countries (2000–2024).
2. **Data Cleaning and Preparation:** Python (Pandas) used to merge datasets, clean missing values, and standardize units.

3. **Data Warehouse Design:** The DW follows a star schema:

- factSolar
- dimCountry
- dimYear
- dimIncome

4. **SQL Analysis:** Used to extract insights on solar growth, emissions, energy shares, and trends.

5. **Visualization and Forecasting:** Python models used to visualize trends and forecast global solar electricity.

6. **GSTI Calculation:** GSTI computed to measure long-term global solar progress.

1.5 Timeline

- Week 1–2: Data collection
- Week 3–4: Data cleaning & merging
- Week 5: Data warehouse design
- Week 6–7: SQL analytics
- Week 8: Visualizations & forecasting
- Week 9: GSTI computation
- Week 10: Final report

1.6 Expected Outcomes

- A cleaned dataset for 66 countries.
- A complete data warehouse.
- SQL-driven global solar insights.
- Solar electricity forecast to 2035.
- GSTI index and interpretation.
- Final report and presentation.

2 Solar Insights

2.1 Global Solar Growth (2000–2024)

Solar energy has shown exponential growth, driven by:

- decreasing solar PV costs,
- government incentives,
- increased energy demand,
- improvements in renewable technology.

Countries such as China, the United States, India, and Germany have led global installation capacity. Many emerging economies in Africa and South America show rapid recent growth.

2.2 Electricity Production from Solar PV

Solar PV electricity output was analyzed across all 66 countries. Key insights include:

- Drastic increase in annual solar electricity generation after 2010.
- High correlation between GDP per capita and solar adoption.
- Faster PV adoption rates in high-income and upper-middle-income countries.

2.3 Solar PV Output Efficiency

Solar PV performance varies by:

- solar irradiance,
- temperature conditions,
- panel efficiency,
- geographical location,
- maintenance quality.

Countries with high irradiance (e.g., MENA region, Australia, Chile) show higher efficiency potential.

2.4 Theoretical vs Technical Solar Potential

- **Theoretical Potential:** Total solar energy that reaches Earth's surface. Almost unlimited but not practically usable.
- **Technical Potential:** Amount of solar energy that can be harvested using available technology and land. Depends on:
 - land availability,
 - topography,
 - solar radiation,
 - technology efficiency.

Countries with vast deserts (Egypt, Saudi Arabia, Australia) have extremely high technical potential.

2.5 Top Countries by Solar Performance

Analysis includes:

- top 10 countries in solar growth (TWh/year),
- countries with highest solar share of electricity,
- correlation between solar adoption and CO₂ reduction,
- solar investment trends.

Visualizations (figures) will be inserted in the final report.